Rebuilding an Equity Volatility Surface from Public Data: Reliability, and a Pre-Registered Look at the Variance Risk Premium

Yasanji Ratnaike
Independent Researcher

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Abstract

This paper rebuilds an equity volatility surface from free, public option data, asks how far it can be trusted, and trades the variance risk premium against it as a worked example. From raw quotes through cleaning, forward extraction, implied-volatility inversion and SVI calibration, the pipeline recovers a known surface exactly on synthetic inputs, so the errors that matter in practice are the data's, concentrated in the wings and the short end. The premium is visible without any model: over 2005 to 2026, VIX exceeded subsequent realised volatility on 85% of days. A short delta-hedged straddle harvests it, with the daily profit and loss attributed to the greeks. The strategy is profitable in a pre-registered development and hold-out split, but the figures are an upper bound, not an edge, resting on VIX as the implied level, a daily rehedge, and costs that omit the real frictions. The components are all established; the contribution is the reliability audit and the honest, pre-registered accounting.

1 Introduction

A volatility surface is the object an options desk reads before it does anything else. It gives an implied volatility for every strike and every expiry, including the great majority that never trade, and it is the input to pricing, to hedging, and to any view on whether options are cheap or dear. Building one is a standard exercise. Building one from data you can actually get hold of, and being honest about how far it can be trusted, is less standard, and it is the part that teaches you something.

This paper does two things. It rebuilds a surface end to end from public data and tests how reliable the result is. It then trades the simplest volatility position against that surface, a short delta-hedged straddle, and attributes the profit and loss to the greeks, so that the source of every pound has a number behind it.

The intent is not to present a strategy, and it is not to present a new method. Every component used here is established, and the surface-building has been done in public before; the next section sets that out. What the paper offers is narrower, and I hope more honest: a pipeline that reproduces faithfully, a way of measuring how far its output can be trusted, and a worked example of the variance risk premium reported under a pre-registered, upper-bound frame. In that last respect it continues Ratnaike (2026): rules fixed in advance, a development and hold-out split, and a willingness to state the limits plainly.

2 Related work

The components used here are established, and the account that follows takes them as given.

The variance risk premium, the tendency of implied volatility to exceed subsequent realised volatility, is documented by Bakshi and Kapadia (2003), whose delta-hedged option gains are close to the object traded in Section 6, and by Carr and Wu (2009); Coval and Shumway (2001) and Bollerslev,

Tauchen and Zhou (2009) sit in the same line. The SVI parameterisation is due to Gatheral (2004), and the butterfly and calendar conditions imposed on it are those of Gatheral and Jacquier (2014). De-Americanisation, pricing an American quote on a tree before inverting it, is studied directly by Burkovska, Glau, Mahlstedt and Wohlmuth (2018). The thirty-day variance-swap reconstruction and its tie to VIX follow Demeterfi, Derman, Kamal and Zou (1999) and the model-free result of Britten-Jones and Neuberger (2000), which also underlies the CBOE VIX construction. The discipline applied to the backtest, pre-registration with a development and hold-out split, follows the backtest-overfitting literature of Bailey and López de Prado (2014) and Harvey, Liu and Zhu (2016), and continues Ratnaike (2026).

The surface-building itself has been done in public many times. Several open-source projects fit an SVI or comparable surface to a free Yahoo Finance option chain, compute greeks, run butterfly and calendar checks, and present the result as an interactive plot. This paper does not improve on that engineering. Its contribution is narrower: it treats the reliability of a public-data surface as something to be measured, through an internal round-trip test and an external comparison with VIX, and it reports the variance-risk-premium example under a pre-registered, upper-bound frame, so the numbers read as a ceiling. The aim is an explicit account of what a surface from public data can and cannot support, which the existing implementations mostly leave open.

3 Data

The option chains come from Yahoo Finance. For a single underlying the pipeline collects every listed option with its strike, expiry, bid, ask, last price, volume and open interest, alongside the spot at the same moment. Index options are taken on SPX, which is European and cash-settled; the American case is handled on SPY, discussed below. Interest rates come from the Treasury curve. Dividends and borrow are not sourced directly; the option prices are allowed to imply them, through the forward.

Everything is held in a point-in-time store. Rows are only added, never overwritten, and each records when it was collected, so the surface on any past date can be rebuilt as it stood. Without that, a bug fixed today silently changes what you thought you saw last week.

The quotes are delayed and retail. Spreads widen a long way from the money, so the data is good enough to learn the mechanics and not good enough to trade on. That is treated as part of the exercise: the cleaning a poor feed forces on you is the same cleaning a good feed needs, with more of it on show. The VIX complex is used later as an independent benchmark, since it is constructed from the same kind of quotes by a published method.

4 Building the surface

A raw chain is filtered before any model sees it. Quotes with a non-positive or crossed bid and ask go first, since a zero bid carries no information. Quotes whose spread is wide against their mid come out next, as stale or illiquid. On each strike the out-of-the-money side is kept, where the options are most liquid and, for American options, carry the least early-exercise premium. A final filter looks at shape: call prices should fall as the strike rises and the curve should be convex, and a quote that breaks that has usually mis-printed. Every rule logs how many quotes it removed.

Two numbers come before any volatility. Using put-call parity across the strikes that have both a call and a put, the forward and the discount factor are backed out for each expiry by a single regression. That one step folds the rate, the dividends and the borrow into one forward, so no separate dividend

feed is needed.

Each clean price is inverted to its Black-Scholes implied volatility. There is no closed form, so it is a numerical search; because price rises monotonically with volatility, the root is unique and a bracketed solver finds it reliably.

Index options are European, but single-name and ETF options such as SPY are American and can be exercised early, which puts an early-exercise premium into the price. Inverting a European formula on an American price would push that premium into the implied volatility and bias it. Each American quote is instead priced on a Cox-Ross-Rubinstein binomial tree that allows early exercise, and the volatility that matches the market price on that tree is solved for. This de-Americanisation keeps the two regimes on the same footing.

The cloud of implied points is fitted one expiry at a time with an SVI parameterisation of total variance, weighted towards the liquid strikes near the money. Within each expiry the fit is constrained to imply no butterfly arbitrage; across expiries total variance is kept from falling as maturity lengthens, which would be a calendar arbitrage. What comes out is a continuous surface, an implied volatility for any strike and expiry.

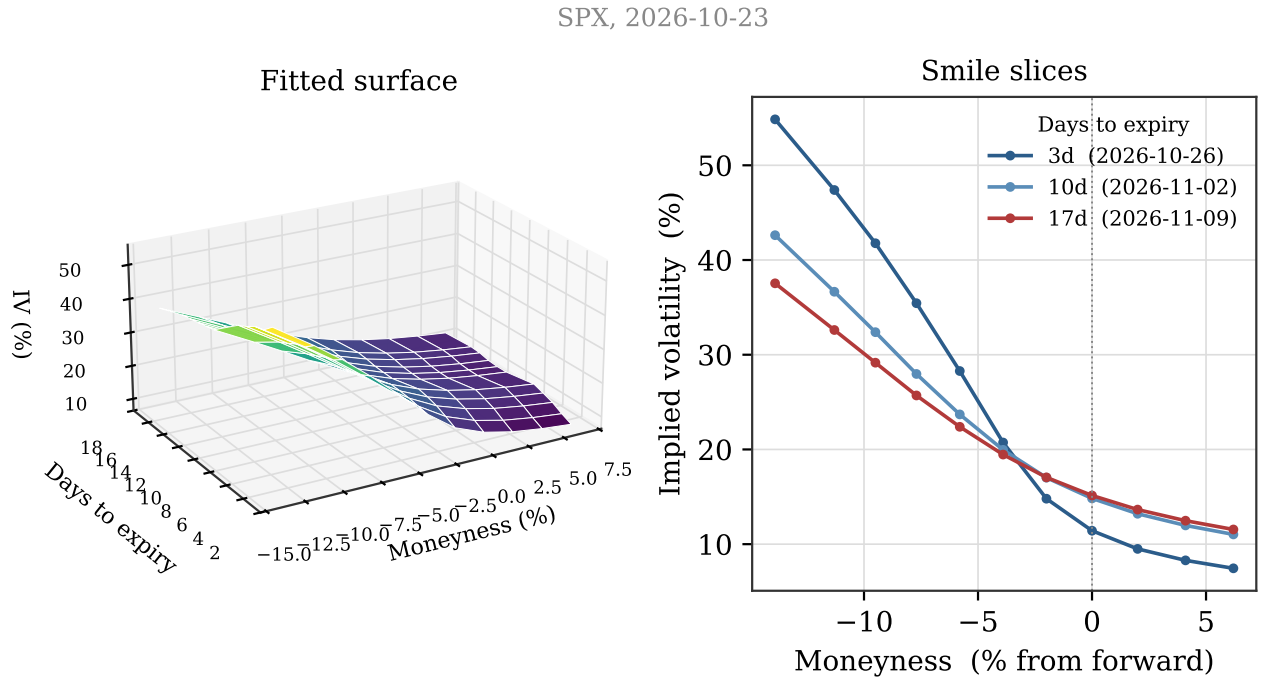


Figure 1: The fitted SPX surface (left) and three of its smile slices (right), on a single snapshot. Volatility rises steeply into the downside strikes, the equity skew, and flattens above the money; the downside slope is steepest at the shortest expiry.

5 How reliable is it?

Reliability is tested two ways: internally, does the pipeline recover what it should, and externally, does its output agree with an independent construction.

The internal test generates option prices from a surface chosen in advance, then runs them back through the whole pipeline to see whether it recovers the inputs. It does, exactly. Across maturities from one month to one year the forward and the discount factor are recovered to within rounding, and the

worst implied-volatility recovery error across every slice is 0.0000 volatility points (Table 1). The fitted surface is butterfly and calendar arbitrage-free with no calendar repairs needed, and the greeks satisfy the internal identity that call delta minus put delta equals $\exp(-qT)$ at every maturity. The estimator is therefore unbiased on clean inputs. Whatever error appears in practice is a data problem, which is exactly what the internal test is there to establish.

Table 1: Round-trip recovery on synthetic inputs. Prices are generated from a known surface and run back through the whole pipeline. The recovered forward and discount factor match the known inputs to the digits shown, and the worst implied-volatility error across all strikes is reported.

Maturity (yr)	Forward (recovered)	Discount (recovered)	Max vol error (vol pts)	Arbitrage
0.08	601.20	0.99681	0.0000	free
0.25	603.76	0.99005	0.0000	free
0.50	607.55	0.98020	0.0000	free
1.00	615.19	0.96079	0.0000	free

Recovered values equal the known inputs to the digits shown. Call delta minus put delta equals $\exp(-qT)$ at every maturity, to machine precision.

The external check rebuilds the thirty-day variance-swap volatility from the whole fitted smile, by the standard replication over out-of-the-money strikes, and compares it to VIX on the same day. VIX is a fair benchmark precisely because it is built the same way from the same kind of data, so close agreement is evidence that the reconstruction is faithful where the market is liquid, and divergence flags where it is not. In the daily tool this comparison runs as a standing check on each rebuilt surface. A full tracking-error series against VIX over the sample needs a stored history of surfaces rather than the single snapshot used here, and is left to the follow-up that also takes up tradability.

The round-trip is clean because its inputs are clean; the live feed is not. The feed strains in two places. The wings, far from the money, are where spreads are widest and quotes stalest, and where the fitted tails are least trustworthy. The very short end is noisy: a three-day expiry can show a deep out-of-the-money put volatility far above its neighbours on a single bad print. The model does not cause these. They are limits of the data, and they fall exactly where a short-volatility position carries its tail risk, which matters for what follows.

6 The variance risk premium, under honest testing

A surface is only interesting once something is traded against it. The simplest honest volatility position is a short one-month at-the-money straddle, delta-hedged each day, so that it carries volatility and not direction. It earns money when the market realises less volatility than was implied, and loses when it realises more.

Before any model is fitted, the premium is visible in the data itself. Over 2005 to 2026, across 5,363 trading days, VIX exceeded the subsequent twenty-one-day realised volatility of the S&P 500 on 84.6% of days, with a mean gap of 3.45 volatility points (Figure 2). That gap is the variance risk premium, and it is visible directly in the data without fitting anything. A short-volatility position is a way of harvesting it.

The point of the exercise is the accounting. Each day's profit and loss is broken into its greeks by a second-order expansion, so the question "where did the money come from" has a numerical answer. Over the full sample the decomposition is dominated by two large and opposing terms: theta contributes $+24.9$ million and gamma -17.1 million, in the arbitrary volatility-point units the book is sized in

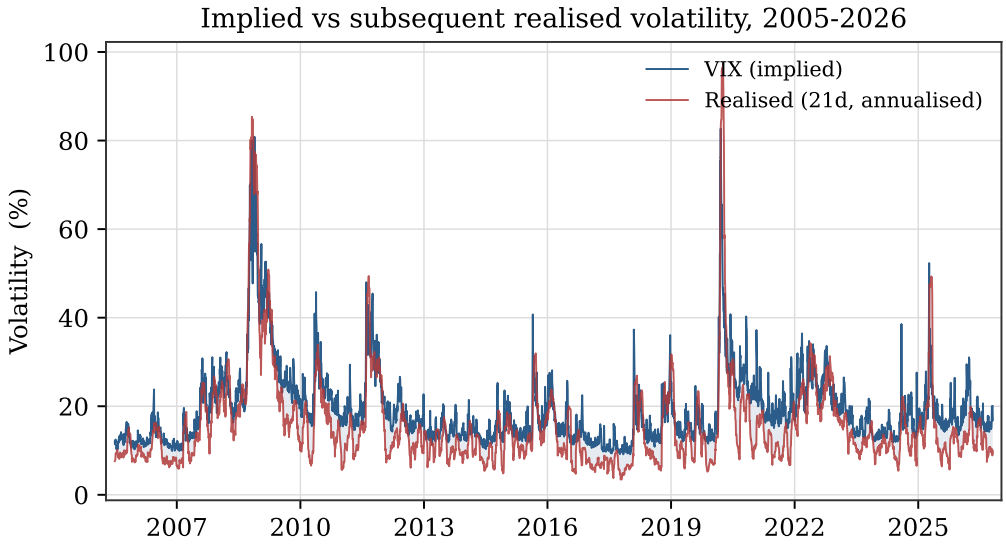


Figure 2: VIX against the subsequent 21-day realised volatility of the S&P 500, 2005–2026. The shaded area marks days on which implied exceeded realised. VIX sits above realised on 84.6% of days, with a mean gap of 3.45 volatility points. The crossings are the crises, where realised briefly overtakes implied.

(Table 2). Their net, +7.7 million, is the premium showing through. Vega, vanna and volga are each under 1.5 million, as expected for a near-the-money delta-hedged straddle; a residual of +0.8 million catches what the expansion misses, and costs take off 0.6 million. The terms reconcile to the reported net to the penny, which is the test that the attribution captures the whole of the profit and loss.

Table 2: Greek attribution of cumulative P&L, full sample, in the vol-target sizing units ($m). The two large terms, theta and gamma, pull against each other; their sum is the premium.

Component	Contribution ($m)
Theta	+24.88
Gamma	−17.14
Theta + Gamma	+7.74
Vega	−0.55
Vanna	+1.40
Volga	−0.55
Residual	+0.76
Costs	−0.62
Net	+8.19

The mechanism behind the two large terms is the delta-hedged identity: a hedged short option makes roughly half of gamma times the squared move, set against the time value it collected. A one-month straddle at sixteen percent implied is priced for a move of about one percent a day. On a quieter day the theta beats the gamma and the day is green; on a larger day gamma takes over and the day is red, and because gamma scales with the square of the move, the red days are the ones that count.

The rules and the development and hold-out split were fixed before the data was seen. On the development period the strategy returns a Sharpe ratio of 1.67, with a Newey-West t-statistic of 6.9; on the hold-out it returns 1.51, with a t-statistic of 5.0 (Table 3). Both periods are positive in each half.

Those numbers are not evidence of a tradable edge, for three reasons that point the same way. First, the

Table 3: Pre-registered results. The development and hold-out split and the trading rules were fixed before the data was seen. Drawdowns are in the vol-target sizing units.

Period	Sharpe	Newey-West t	Max drawdown	% days short
Development	1.67	6.9	−519k	94.2
Hold-out	1.51	5.0	−699k	92.9
Full sample	1.60	8.3	−699k	93.6

variance risk premium is one of the best-documented premia in the literature, so a positive development and a positive hold-out are what the prior predicts; they show the pipeline reproduces a known effect. Discovering one would be a different and much stronger claim. Second, the Sharpe is an upper bound. The implied level is VIX, a published index used in place of an executable surface, the hedge is re-struck once a day at the close, the only cost charged is a tenth of a volatility point a side, and nothing is deducted for the wide bid-ask in the wings, for overnight gaps, or for the margin and financing a short-volatility book consumes. Each of those omissions flatters the result, and they are precisely the frictions that separate a backtest from a trade. Third, the tail is real and large. The worst drawdowns in the sample are −699,000 in the COVID crash of early 2020, −519,000 in the 2008 crisis, and −287,000 in the tariff sell-off of April 2025, all in the same sizing units (Figure 3). A short-volatility book's losses cluster, arrive fast, and dwarf the daily premium it collects the rest of the time. The premium is real; the figures here are its ceiling, well above what the trade would actually return.

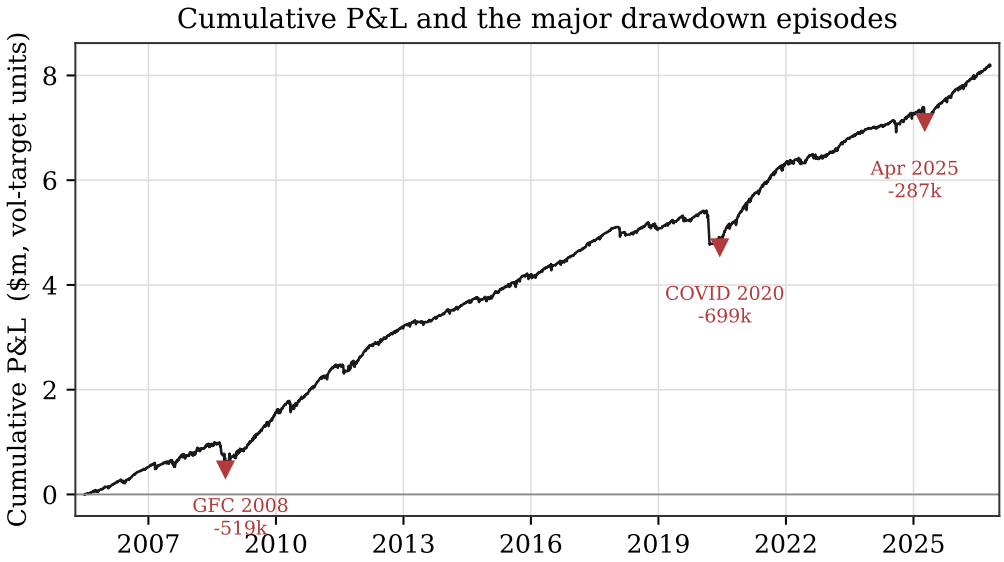


Figure 3: Cumulative P&L of the delta-hedged short straddle in the vol-target sizing units, with the major drawdown episodes marked. The steady climb is the premium; the sharp steps down are where a short-volatility book is tested.

7 Limitations

Every result above rests on assumptions worth stating plainly. The quotes are delayed retail data, worst in the wings where the tails of the surface live. In the backtest VIX stands in for the implied level, because a free history of option surfaces does not exist, and that substitution rests on the VIX agreement check, since a traded surface is not available. The position is sized to a constant volatility

exposure, so the figures describe how the strategy behaves and are not a return on capital. The hedge is re-struck once a day at the close, which is cleaner than anything achievable in practice. And a delta-hedged short-volatility position carries genuine tail risk that a Sharpe ratio, computed over mostly quiet days, understates by construction.

8 Conclusion

The exercise set out to rebuild the volatility machinery from data anyone can get, and to be honest about the result. Two things come out of it. The surface can be reconstructed faithfully: on clean inputs the pipeline recovers the inputs exactly and the output is arbitrage-free, so the limits in practice are data limits, which the paper locates in the wings and the short end. And the variance risk premium is plainly visible in twenty-one years of free data, with an attribution that breaks the profit and loss into its terms, so that no part of it is left unexplained.

What the exercise cannot show is tradability, and it does not try to. Tradability lives in the frictions the free data and the daily idealisation leave out, and those are the honest subject of any follow-up. The framework, the reliability tests, and the attribution are reusable for that, and for any other position taken against the surface.

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Open-source implied-volatility-surface implementations on public data, for example `github.com/ParBproject/options-vol-s` and `github.com/Achintya1311/implied-vol-surface` (accessed October 2026).

A Key relationships

Black-76 (forward form). A European call is priced on the forward F as

$$C = D\,[\,F\,N(d_1) - K\,N(d_2)\,], \qquad d_{1,2} = \frac{\ln(F/K) \pm \tfrac{1}{2}\sigma^2 T}{\sigma\sqrt{T}},$$

with D the discount factor. Price is strictly increasing in σ, which guarantees a unique implied volatility.

Put-call parity for the forward. For each expiry, $C - P = D\,(F - K)$, so a regression of $C - P$ on K across paired strikes gives the discount factor from the slope and the forward from the intercept, folding rates, dividends and borrow into F.

SVI (raw). Total variance in log-moneyness k is

$$w(k) = a + b\,\big[\rho\,(k - m) + \sqrt{(k - m)^2 + \sigma^2}\,\big].$$

Butterfly arbitrage is excluded by requiring Gatheral's $g(k) \geq 0$; calendar arbitrage by requiring $w(k)$ non-decreasing in maturity.

P&L attribution. The daily change is expanded as

$$\Delta V \approx \Theta\,\Delta t + \Delta\,\Delta S + \tfrac{1}{2}\Gamma\,\Delta S^2 + \mathcal{V}\,\Delta\sigma + \text{Vanna}\,\Delta S\,\Delta\sigma + \tfrac{1}{2}\text{Volga}\,\Delta\sigma^2,$$

with the delta term removed by the hedge and a residual for the higher-order terms. For the hedged straddle this reduces to roughly $\tfrac{1}{2}\Gamma S^2(\sigma^2_{\text{real}} - \sigma^2_{\text{imp}})$.

Variance-swap / VIX replication. The fair thirty-day variance is rebuilt from a strip of out-of-the-money options weighted by $1/K^2$, the same construction VIX uses, which is why VIX is a fair independent check on the fitted surface.